Decent Coaching Classes

X Semi — Mind Your Math - I

Answer Key

Q.1 (A) Choose the correct alternative

1) b) $ax + by + c = 0$
2) b) no solution
3) b) $ax^2 + bx + c = 0, a \neq 0$
4) b) real and equal

Q.1 (B) Solve the following

1) $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$
2) $D = a_1b_2 - a_2b_1$
3) Any valid example, e.g. $x^2 + 5x + 6 = 0$
4) $b^2 - 4ac > 0$

Q.2 (A) Activities

1) $3x + 2y = 12$; $x - 2y = 4$

Adding: $4x = 16 \Rightarrow x = 4$

Substituting in (II): $4 - 2y = 4 \Rightarrow -2y = 0 \Rightarrow y = 0$

Solution: $(x, y) = (4, 0)$

2) $x^2 + 7x + 10 = 0$

$x^2 + 5x + 2x + 10 = 0 \Rightarrow x(x + 5) + 2(x + 5) = 0 \Rightarrow (x + 5)(x + 2) = 0$

$x = -5$ or $x = -2$

3) $2x^2 + 5x + 3 = 0$

$a = 2, b = 5, c = 3$

$b^2 - 4ac = 25 - 24 = 1$

Since discriminant > 0 and a perfect square, roots are real, unequal and rational

Q.2 (B) Solve the following

1) $2x + 3y = 11$; $x - 2y = -3 \rightarrow x = 1, y = 3$
2) $x + y = 4$; $x - y = 2 \rightarrow D = -2, Dx = -4, Dy = -2 \rightarrow x = 3, y = 1$
3) $x^2 - 5x + 6 = 0 \rightarrow (x - 2)(x - 3) = 0 \rightarrow x = 2$ or $x = 3$
4) $x^2 + 2x - 15 = 0 \rightarrow (x + 5)(x - 3) = 0 \rightarrow x = -5$ or $x = 3$
5) $2x^2 - 4x + 3 = 0 \rightarrow a = 2, b = -4, c = 3 \rightarrow D = 16 - 24 = -8$ (roots not real)

Q.3 Solve the following

1) $x^2 + 6x + 5 = 0 \rightarrow (x + 3)^2 = 4 \rightarrow x + 3 = \pm 2 \rightarrow x = -1$ or $x = -5$
2) $x^2 - 3x - 4 = 0 \rightarrow x = (3 \pm \sqrt{25})/2 = (3 \pm 5)/2 \rightarrow x = 4$ or $x = -1$
3) Let numbers be x and y. $x + y = 9$; $x - y = 3 \rightarrow x = 6, y = 3$

Q.4 Solve the following

1) Let breadth = x cm, length = (x + 5) cm

$x(x + 5) = 84 \rightarrow x^2 + 5x - 84 = 0 \rightarrow (x + 12)(x - 7) = 0 \rightarrow x = 7$ (rejecting negative)

Breadth = 7 cm, Length = 12 cm

2) Let numbers be x and x + 1

$x^2 + (x + 1)^2 = 61 \rightarrow 2x^2 + 2x + 1 = 61 \rightarrow 2x^2 + 2x - 60 = 0 \rightarrow x^2 + x - 30 = 0$

$(x + 6)(x - 5) = 0 \rightarrow x = 5$ (rejecting negative)

Numbers are 5 and 6